
Wisest

Curriculum Database

Topics & Subtopics Registry

GCSE (Foundation & Higher) · A-level Mathematics
A-level Further Mathematics

Every subtopic is a record with a stable primary-key id,
name, tier and coverage description.

July 3, 2026

Seed registry for the Wisest question-bank pipeline.
Each id is the key against which a 55-question bank is generated and verified.

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Schema

Each row below is one **subtopic record**. Fields:

id — primary key. Stable dot-slug, globally unique, never reused. This is the identifier every generation agent, ledger entry and bank file references.

subtopic — human-readable name.

tier (GCSE only) — **F** = examined on both Foundation and Higher tiers; **H** = Higher tier only.

description — one-line summary of the mathematics covered; the generation brief for that record.

Key grammar.

GCSE: `gcse.\textit{strand}.\textit{subtopic}`

A-level Maths: `al.\textit{year}.\textit{strand}.\textit{subtopic}` (year ∈ y1, y2; strand ∈ pure, stats, mech)

Further Maths: `fm.\textit{year}.\textit{strand}.\textit{subtopic}` (pure = Core Pure; stats = Further Statistics; mech = Further Mechanics)

The PDF groups records by topic for readability; in the production canonical JSON these topics are an explicit nesting level between strand and subtopic. Tiering follows the DfE/Edexcel 1MA1 typographic convention (bold = Higher only). A-level year split follows the standard AS/A2 teaching split; Further Maths applied content follows the Edexcel Further Statistics / Further Mechanics option route.

Registry summary

Qualification	Strand / module	Records
GCSE (123)	Number	25
	Algebra	37
	Ratio, proportion & rates of change	15
	Geometry and measures	29
	Probability	9
	Statistics	8
	<i>Foundation-tier (F)</i>	91
	<i>Higher-only (H)</i>	32
A-level Maths (83)	Year 1 Pure	27
	Year 1 Statistics	9
	Year 1 Mechanics	6
	Year 2 Pure	31
	Year 2 Statistics	5
	Year 2 Mechanics	5
Further Maths (43)	Y1 Pure (Core Pure 1)	9
	Y1 Statistics (FS1)	6
	Y1 Mechanics (FM1)	4
	Y2 Pure (Core Pure 2)	10
	Y2 Statistics (FS2)	7
	Y2 Mechanics (FM2)	7
Total		249

249 subtopic records × 55 questions each = **13,695 questions** at full generation. GCSE Foundation banks draw from the 91 F-tagged records; Higher banks from all 123. Counts are the current decomposition and may shift by a few records as subtopics are split or merged during review — the id of an existing record is never changed once questions reference it.

Part I — GCSE Mathematics (Foundation & Higher)

Key prefix `gcse.` — six strands. F = both tiers; H = Higher only.

1 Number

1.1 Structure and calculation

id	subtopic	tier	description
<code>gcse.number.ordering</code>	Ordering and comparing numbers	F	Order integers, decimals and fractions; use $=, \neq, <, >, \leq, \geq$.
<code>gcse.number.four-operations</code>	Four operations and place value	F	Formal written methods for integers, decimals, fractions and mixed numbers; place value.
<code>gcse.number.negatives</code>	Negative numbers	F	Calculating with directed numbers across all four operations, in and out of context.
<code>gcse.number.order-of-operations</code>	Priority of operations	F	Brackets, powers, roots, reciprocals; inverse operations and cancellation.
<code>gcse.number.factors-multiples</code>	Factors, multiples and primes	F	Primes, factors, multiples, common factors/multiples, HCF and LCM by listing.
<code>gcse.number.prime-factorisation</code>	Prime factorisation	F	Product of primes in index form; HCF and LCM via factorisation and Venn methods.
<code>gcse.number.systematic-listing</code>	Systematic listing	F	Enumerating outcomes and arrangements without omission or repetition.
<code>gcse.number.product-rule</code>	Product rule for counting	H	m ways then n ways gives $m \times n$; applied counting problems.
<code>gcse.number.powers-roots</code>	Powers and roots	F	Integer powers, square/cube/higher roots; recognising powers of 2,3,4,5.
<code>gcse.number.integer-indices</code>	Integer indices	F	Index laws for products, quotients, powers of powers; zero and negative indices.
<code>gcse.number.fractional-indices</code>	Fractional indices	H	Evaluating and manipulating unit and non-unit fractional powers.
<code>gcse.number.estimate-powers</code>	Estimating powers and roots	H	Estimating powers and roots by bracketing between known values.
<code>gcse.number.exact-calculation</code>	Exact calculation	F	Calculating exactly with fractions and multiples of π .
<code>gcse.number.surds</code>	Surds	H	Simplifying surds, expanding surd brackets, rationalising denominators.
<code>gcse.number.standard-form</code>	Standard form	F	Convert to/from $A \times 10^n$, $1 \leq A < 10$; four operations in standard form.

1.2 Fractions, decimals and percentages

id	subtopic	tier	description
<code>gcse.number.fraction-arithmetic</code>	Fraction arithmetic	F	Four operations on proper/improper fractions and mixed numbers.
<code>gcse.number.fdp-equivalence</code>	FDP equivalence	F	Converting between decimals, fractions and percentages both ways.
<code>gcse.number.recurring-decimals</code>	Recurring decimals	H	Proving recurring decimals equal exact fractions; converting both ways.

continued

id	subtopic	tier	description
gcse.number.fractions-of-amounts	Fractions of amounts	F	Fractions as operators; fractions within ratio problems.
gcse.number.percentage-operators	Percentages as operators	F	Percentage of an amount by multiplier, with and without a calculator.

1.3 Measures and accuracy

id	subtopic	tier	description
gcse.number.units	Standard and compound units	F	Mass, length, time, money; converting units and compound measures.
gcse.number.estimation	Estimation and checking	F	Approximating to 1 s.f.; checking answers from technology.
gcse.number.rounding	Rounding	F	Rounding to decimal places and significant figures; appropriate accuracy.
gcse.number.error-intervals	Error intervals	F	Inequality notation for the interval after rounding or truncation.
gcse.number.bounds	Upper and lower bounds	H	Bounds of compound calculations; guaranteeing max/min values.

2 Algebra

2.1 Notation, vocabulary and manipulation

id	subtopic	tier	description
gcse.algebra.notation	Algebraic notation	F	Conventions ab , $3y$, a^2 , $\frac{a}{b}$; interpreting expressions.
gcse.algebra.substitution	Substitution	F	Substituting values into expressions and scientific formulae.
gcse.algebra.vocabulary	Algebraic vocabulary	F	Distinguishing expressions, equations, formulae, identities, terms, factors.
gcse.algebra.simplifying	Simplifying and single brackets	F	Collecting like terms; expanding a single bracket; common factors.
gcse.algebra.expanding-binomials	Expanding binomials	F	Products of two binomials; Higher extends to more than two.
gcse.algebra.factorising-quadratics	Factorising quadratics	F	Factorising $x^2 + bx + c$ and the difference of two squares.
gcse.algebra.factorising-ax2	Factorising $ax^2 + bx + c$	H	General quadratics with leading coefficient greater than one.
gcse.algebra.algebraic-fractions	Algebraic fractions	H	Simplifying and four operations on algebraic fractions.
gcse.algebra.index-laws	Index laws in algebra	F	Simplifying algebraic expressions using the laws of indices.
gcse.algebra.rearranging	Rearranging formulae	F	Changing the subject, including where the subject appears twice.
gcse.algebra.identities-proof	Identities and argument	F	Showing equivalence; algebraic argument; formal proof at Higher.
gcse.algebra.functions	Functions	H	Function notation; inverse and composite functions.

2.2 Graphs

id	subtopic	tier	description
gcse.algebra.coordinates	Coordinates	F	All four quadrants; midpoints and coordinate problems.
gcse.algebra.straight-line-graphs	Straight-line graphs	F	Plotting $y = mx + c$; equations from gradient/point or two points.
gcse.algebra.gradient-intercept	Gradient and intercepts	F	Interpreting gradients and intercepts graphically and algebraically.
gcse.algebra.parallel-perp	Parallel and perpendicular lines	F	Parallel via equal gradients (F); perpendicular $m_1 m_2 = -1$ (H).
gcse.algebra.quadratic-graphs	Quadratic graphs	F	Parabolas; roots, intercepts, turning points from graphs.
gcse.algebra.turning-points	Turning points by completing square	H	Deducing turning points algebraically and sketching.
gcse.algebra.cubic-reciprocal-graphs	Cubic and reciprocal graphs	F	Recognising and sketching $y = x^3$ shapes and $y = \frac{1}{x}$.
gcse.algebra.exp-trig-graphs	Exponential and trig graphs	H	Graphs of $y = k^x$ and of sin, cos, tan for any angle.
gcse.algebra.graph-transformations	Transformations of graphs	H	Translations and reflections of $f(x)$.
gcse.algebra.real-life-graphs	Real-life graphs	F	Interpreting context graphs including distance–time and speed–time.
gcse.algebra.gradients-areas	Gradients and areas under curves	H	Rates via tangents and areas via trapezia; kinematic reading.
gcse.algebra.circle-equation	Equation of a circle	H	$x^2 + y^2 = r^2$; tangent to a circle at a point.

2.3 Solving equations and inequalities

id	subtopic	tier	description
gcse.algebra.linear-equations	Linear equations	F	One unknown, unknowns both sides, fractional coefficients.
gcse.algebra.quadratics-factorising	Quadratics by factorising	F	Solving by factorisation; approximate solutions from graphs.
gcse.algebra.quadratic-formula	Completing square and formula	H	Solving by completing the square and the quadratic formula.
gcse.algebra.simultaneous-linear	Simultaneous equations (linear)	F	Two linear equations, algebraically and graphically.
gcse.algebra.simultaneous-quadratic	Simultaneous (linear/quadratic)	H	One linear and one quadratic; algebraic and graphical.
gcse.algebra.iteration	Iteration	H	Approximate solutions by iterative formulae; roots in intervals.
gcse.algebra.forming-equations	Forming and solving	F	Translating situations into equations and interpreting solutions.
gcse.algebra.linear-inequalities	Linear inequalities	F	Solving in one variable; solution sets on number lines.

continued

id	subtopic	tier	description
gcse.algebra. quadratic-inequalities	Quadratic & 2-variable inequalities	H	Quadratic inequalities; regions in two variables; set notation.

2.4 Sequences

id	subtopic	tier	description
gcse.algebra. generating-sequences	Generating sequences	F	Terms from term-to-term and position-to-term rules.
gcse.algebra. special-sequences	Special sequences	F	Triangular, square, cube, arithmetic, Fibonacci, quadratic, geometric.
gcse.algebra.nth-term-linear	n th term of linear sequences	F	Deducing and using the n th term of arithmetic sequences.
gcse.algebra. nth-term-quadratic	n th term of quadratic sequences	H	n th term of quadratic sequences; surd-ratio geometric sequences.

3 Ratio, Proportion and Rates of Change

3.1 Ratio

id	subtopic	tier	description
gcse.ratio.notation	Ratio notation and simplifying	F	Writing, simplifying and using ratios in simplest form.
gcse.ratio.dividing	Dividing in a ratio	F	Part:part and part:whole division; expressing divisions as ratios.
gcse.ratio.problem-solving	Ratio problem solving	F	Conversion, comparison, scaling, mixing; combining ratios.
gcse.ratio.fractions-linear	Ratios, fractions and linear functions	F	Multiplicative relationships as ratios/fractions; link to linear graphs.

3.2 Percentages and proportion

id	subtopic	tier	description
gcse.ratio.percentage-change	Percentage change	F	Increase/decrease with multipliers; one quantity as a percentage of another.
gcse.ratio. reverse-percentages	Reverse percentages	F	Finding original values after a percentage change.
gcse.ratio.interest	Simple and compound interest	F	Simple interest; compound growth and decay problems.
gcse.ratio.iterative-growth	Iterative growth processes	H	General iterative processes for repeated proportional change.
gcse.ratio.direct-inverse	Direct and inverse proportion	F	Numerical and graphical problems; recognising proportion graphs.
gcse.ratio. algebraic-proportion	Algebraic proportion	H	Building and using $y = kx$, $y = \frac{k}{x}$ and power models.

3.3 Rates and compound measures

id	subtopic	tier	description
gcse.ratio.unit-conversion	Unit conversion	F	Converting standard units in numerical and algebraic contexts.
gcse.ratio.scale	Scale factors, diagrams and maps	F	Scale drawings and map scales; lengths and areas under scaling.
gcse.ratio.compound-measures	Compound measures	F	Speed, density, pressure, rates of pay; converting compound units.
gcse.ratio.rates-on-graphs	Rates of change on graphs	F	Gradient as rate of change; direct/inverse proportion graphs.
gcse.ratio.instantaneous-rate	Instantaneous rate of change	H	Gradients of chords and tangents; average vs instantaneous rate.

4 Geometry and Measures

4.1 Properties and constructions

id	subtopic	tier	description
gcse.geometry.notation	Notation and drawing	F	Terms for points, lines, planes, polygons; labelling; drawing from description.
gcse.geometry.constructions	Constructions and loci	F	Ruler-and-compass constructions; loci combining conditions.
gcse.geometry.angle-facts	Angle facts	F	Angles at a point, on a line, vertically opposite; angle equations.
gcse.geometry.parallel-angles	Angles in parallel lines	F	Alternate, corresponding and co-interior angles; multi-step reasoning.
gcse.geometry.polygon-angles	Angles in polygons	F	Interior and exterior angles; regular polygon properties.
gcse.geometry.quadrilaterals	Quadrilateral properties	F	Definitions and properties of the special quadrilaterals.
gcse.geometry.congruence	Congruence	F	SSS, SAS, ASA, RHS; simple congruence proofs.
gcse.geometry.similarity	Similarity	F	Similar shapes; length scale factors; similarity arguments.
gcse.geometry.transformations	Transformations	F	Rotation, reflection, translation, enlargement (negative s.f. at Higher).
gcse.geometry.combined-transformations	Combining transformations	H	Single equivalent transformations; invariance.
gcse.geometry.circle-vocabulary	Circle vocabulary	F	Centre, radius, chord, diameter, tangent, arc, sector, segment.
gcse.geometry.circle-theorems	Circle theorems	H	The standard circle theorems, their proofs and problems.
gcse.geometry.3d-plans-elevations	3D solids, plans and elevations	F	Properties of solids; plans and elevations.

4.2 Mensuration and calculation

id	subtopic	tier	description
<code>gcse.geometry.perimeter-area</code>	Perimeter and area	F	Triangles, parallelograms, trapezia and composite shapes.
<code>gcse.geometry.circles-area</code>	Circles: circumference and area	F	$C = 2\pi r$, $A = \pi r^2$, exactly and to given accuracy.
<code>gcse.geometry.arcs-sectors</code>	Arcs and sectors	F	Arc length, sector area and sector angles; sector perimeter.
<code>gcse.geometry.volume-prisms</code>	Volume and surface area of prisms	F	Cuboids, right prisms, cylinders; composite prisms.
<code>gcse.geometry.spheres-cones</code>	Spheres, cones and pyramids	F	Surface area and volume, including composite solids.
<code>gcse.geometry.similar-area-volume</code>	Similar lengths, areas and volumes	H	Length, area and volume scale-factor relationships.
<code>gcse.geometry.bearings</code>	Bearings and scale drawings	F	Three-figure bearings; maps and accurate drawings.

4.3 Pythagoras and trigonometry

id	subtopic	tier	description
<code>gcse.geometry.pythagoras</code>	Pythagoras' theorem	F	Sides in right-angled triangles; distance between coordinates.
<code>gcse.geometry.trig-ratios</code>	Trigonometric ratios	F	Sine, cosine, tangent for sides and angles in right-angled triangles.
<code>gcse.geometry.exact-trig</code>	Exact trigonometric values	F	Exact sin, cos for 0–90° and tan for 0–60°.
<code>gcse.geometry.3d-trig</code>	3D Pythagoras and trigonometry	H	Lengths and angles in three-dimensional figures.
<code>gcse.geometry.sine-cosine-rule</code>	Sine and cosine rules	H	General triangles, including the ambiguous case.
<code>gcse.geometry.triangle-area</code>	Area of a triangle	H	$\frac{1}{2}ab \sin C$ for area, sides or angles of any triangle.

4.4 Vectors

id	subtopic	tier	description
<code>gcse.geometry.vector-notation</code>	Vector notation and translations	F	Column vectors for translations; diagrammatic representation.
<code>gcse.geometry.vector-arithmetic</code>	Vector arithmetic	F	Addition, subtraction and scalar multiplication in diagram and column form.
<code>gcse.geometry.vector-proof</code>	Vector geometry and proof	H	Geometric arguments and proofs; collinearity and midpoints.

5 Probability

id	subtopic	tier	description
<code>gcse.probability.experiments</code>	Experiments and frequency	F	Recording outcomes with tables and frequency trees.
<code>gcse.probability.expected-outcomes</code>	Expected outcomes	F	Randomness, fairness, equally likely events; expected frequency.

continued

id	subtopic	tier	description
gcse.probability. scale-relative-freq	Probability scale and relative frequency	F	The 0–1 scale; relative frequency vs theoretical probability.
gcse.probability. exhaustive-events	Exhaustive and mutually exclusive events	F	Probabilities summing to one; complement rules.
gcse.probability. sampling-convergence	Sampling and convergence	F	Empirical distributions tending to theoretical with sample size.
gcse.probability.sets-venn	Sets and Venn diagrams	F	Enumerating sets and combinations with tables, grids, Venn diagrams.
gcse.probability. possibility-spaces	Possibility spaces	F	Sample spaces for single and combined experiments.
gcse.probability. tree-diagrams	Tree diagrams	F	Independent and dependent combined events; assumptions.
gcse.probability.conditional	Conditional probability	H	Conditional probability via frequencies, two-way tables, trees, Venn.

6 Statistics

id	subtopic	tier	description
gcse.statistics.sampling	Sampling	F	Inferring population properties from samples; limitations.
gcse.statistics. charts-tables	Charts and tables	F	Frequency tables, bar/pie charts, pictograms, vertical line charts.
gcse.statistics.time-series	Time series	F	Tables and line graphs for time series data; describing trends.
gcse.statistics. averages-spread	Averages and spread	F	Mean, median, mode, modal class, range from lists and tables; outliers.
gcse.statistics.comparing	Comparing distributions	F	Comparing data sets using centre and spread.
gcse.statistics. histograms-cf	Histograms and cumulative frequency	H	Unequal-width histograms and frequency density; cumulative frequency.
gcse.statistics. quartiles-boxplots	Quartiles and box plots	H	Quartiles and IQR; drawing, reading and comparing box plots.
gcse.statistics. scatter-correlation	Scatter graphs and correlation	F	Correlation vs causation; lines of best fit; interpolation/extrapolation.

Part II — A-level Mathematics, Year 1 (AS content)

Key prefix `al.y1.` — Pure, Statistics and Mechanics.

7 Year 1 Pure

id	subtopic	description
<code>al.y1.pure.argument-notation</code>	Mathematical argument and notation	Statement structure; implication, equivalence, necessary and sufficient conditions.
<code>al.y1.pure.proof-deduction</code>	Proof by deduction and exhaustion	Direct algebraic proofs; finite-case proof by exhaustion.
<code>al.y1.pure.disproof</code>	Disproof by counterexample	Refuting general claims with a single counterexample.
<code>al.y1.pure.indices-surds</code>	Indices and surds	Rational-exponent index laws; manipulating and rationalising surds.
<code>al.y1.pure.quadratics</code>	Quadratic functions	Completing the square, discriminant, sketching, quadratics in a function of x .
<code>al.y1.pure.simultaneous</code>	Simultaneous equations	Linear/linear and linear/quadratic systems; intersection interpretation.
<code>al.y1.pure.inequalities</code>	Inequalities	Linear and quadratic inequalities; set notation and solution regions.
<code>al.y1.pure.polynomials</code>	Polynomials and factor theorem	Expanding, factorising, dividing polynomials; factor theorem for cubics.
<code>al.y1.pure.curve-sketching</code>	Curve sketching	Polynomial and reciprocal curves; asymptotes; solving via sketches.
<code>al.y1.pure.graph-transformations</code>	Transformations of graphs	Effects of $af(x)$, $f(x) + a$, $f(x + a)$, $f(ax)$ on standard curves.
<code>al.y1.pure.straight-lines</code>	Straight lines	Gradient, midpoint, distance; $ax + by + c = 0$; parallel/perpendicular; modelling.
<code>al.y1.pure.circles</code>	Circles	Centre–radius equation; completing the square; tangent and chord properties.
<code>al.y1.pure.binomial</code>	Binomial expansion	$(a + bx)^n$ for positive integer n ; $\binom{n}{r}$; coefficients and approximations.
<code>al.y1.pure.triangle-trig</code>	Triangle trigonometry	Sine/cosine rules, area formula; bearings and the ambiguous case.
<code>al.y1.pure.trig-graphs</code>	Trigonometric graphs	Graphs and symmetries of $\sin$, $\cos$, $\tan$; exact values; transformations.
<code>al.y1.pure.trig-identities</code>	Identities and equations	$\tan \equiv \frac{\sin}{\cos}$, $\sin^2 + \cos^2 \equiv 1$; solving trig equations in intervals.
<code>al.y1.pure.exponentials</code>	Exponential functions	Graphs of a^x and e^x ; gradient property of e^{kx} .
<code>al.y1.pure.logarithms</code>	Logarithms	Logarithm as inverse; laws of logs; solving $a^x = b$.
<code>al.y1.pure.exp-modelling</code>	Exponential modelling	Growth/decay models; use of $\ln$; reduction to linear log-form.
<code>al.y1.pure.differentiation</code>	First principles and derivatives	Derivative as gradient/rate; differentiating x^n from first principles.
<code>al.y1.pure.tangents-normals</code>	Tangents and normals	Gradients at points; tangent and normal equations.
<code>al.y1.pure.stationary-points</code>	Increasing, decreasing, stationary points	Second-derivative test; sketching gradient functions.
<code>al.y1.pure.optimisation</code>	Optimisation	Setting up and solving practical maximum/minimum problems.
<code>al.y1.pure.integration</code>	Indefinite integration	Antidifferentiation; integrating x^n ; finding c from a point.
<code>al.y1.pure.definite-integration</code>	Definite integration and areas	Definite integrals; areas between curves and the x -axis.

continued

id	subtopic	description
<code>al.y1.pure.vectors</code>	Vector basics	Magnitude, direction, unit vectors; $\mathbf{i}$, $\mathbf{j}$ and column notation.
<code>al.y1.pure.vector-geometry</code>	Vector geometry	Position vectors; distance; ratios and collinearity; mechanics links.

8 Year 1 Statistics

id	subtopic	description
<code>al.y1.stats.sampling</code>	Sampling	Population vs sample; random, systematic, stratified, quota, opportunity methods.
<code>al.y1.stats.location-spread</code>	Measures of location and spread	Mean, median, quartiles, range, IQR, variance, s.d.; linear coding.
<code>al.y1.stats.data-presentation</code>	Data presentation	Box plots, cumulative frequency, histograms; outliers; comparing distributions.
<code>al.y1.stats.regression</code>	Correlation and regression	Scatter diagrams; least-squares lines; interpretation and extrapolation limits.
<code>al.y1.stats.probability</code>	Probability	Venn and tree diagrams; mutually exclusive and independent events.
<code>al.y1.stats.discrete-rv</code>	Discrete random variables	Probability distributions and mass functions; discrete uniform.
<code>al.y1.stats.binomial</code>	The binomial distribution	Conditions; single and cumulative binomial probabilities.
<code>al.y1.stats.hypothesis-testing</code>	Hypothesis testing	Hypotheses, significance, one/two-tailed binomial tests, critical regions.
<code>al.y1.stats.large-data-set</code>	Large data set fluency	Contexts, units, sampling and typical values of the prescribed data set.

9 Year 1 Mechanics

id	subtopic	description
<code>al.y1.mech.modelling</code>	Modelling in mechanics	Modelling assumptions; SI units; scalars and vectors.
<code>al.y1.mech.kinematics-graphs</code>	Kinematics graphs	Displacement–time and velocity–time graphs; gradients and areas.
<code>al.y1.mech.constant-acceleration</code>	Constant acceleration	Deriving and applying <i>suvat</i> ; vertical motion under gravity.
<code>al.y1.mech.variable-acceleration</code>	Variable acceleration	Calculus between displacement, velocity and acceleration in 1D.
<code>al.y1.mech.forces-newton</code>	Forces and Newton's laws	Force diagrams, resultants, $F = ma$; motion in 1D and 2D; weight.
<code>al.y1.mech.connected-particles</code>	Connected particles	Newton's third law; pulleys, tow-bars and lifts.

Part III — A-level Mathematics, Year 2

Key prefix `al.y2.` — Pure, Statistics and Mechanics.

10 Year 2 Pure

id	subtopic	description
<code>al.y2.pure.proof-contradiction</code>	Proof by contradiction	Contradiction arguments; irrationality of $\sqrt{2}$; infinitude of primes.
<code>al.y2.pure.partial-fractions</code>	Algebraic and partial fractions	Simplifying rational expressions; linear and repeated linear factors.
<code>al.y2.pure.functions</code>	Functions and their graphs	Domain/range; composite and inverse functions; the modulus function.
<code>al.y2.pure.further-transformations</code>	Further transformations	Combined transformations and their effect on domain and range.
<code>al.y2.pure.arithmetic-series</code>	Arithmetic sequences and series	n th term, sum formulae, sigma notation; modelling.
<code>al.y2.pure.geometric-series</code>	Geometric sequences and series	n th term, finite sums, convergence, sum to infinity.
<code>al.y2.pure.recurrence</code>	Recurrence and general sequences	Increasing, decreasing, periodic sequences; recurrence relations.
<code>al.y2.pure.binomial-rational</code>	Binomial expansion (rational powers)	$(1+x)^n$ for negative/fractional n ; validity; use with partial fractions.
<code>al.y2.pure.radians</code>	Radians, arcs and sectors	Radian measure; $r\theta$ and $\frac{1}{2}r^2\theta$; solving in radians.
<code>al.y2.pure.small-angles</code>	Small angle approximations	$\sin \theta \approx \theta$, $\tan \theta \approx \theta$, $\cos \theta \approx 1 - \frac{\theta^2}{2}$; validity.
<code>al.y2.pure.reciprocal-inverse-trig</code>	Reciprocal and inverse trig functions	sec, csc, cot and inverse trig functions; associated identities.
<code>al.y2.pure.compound-double-angles</code>	Compound and double angles	Addition and double-angle formulae; proving identities; equations.
<code>al.y2.pure.rsin-form</code>	The form $R \sin(\theta + \alpha)$	Expressing $a \sin \theta + b \cos \theta$ in R form; maxima/minima; modelling.
<code>al.y2.pure.parametric-curves</code>	Parametric curves	Sketching and converting parametric/Cartesian forms via identities.
<code>al.y2.pure.parametric-modelling</code>	Parametric modelling	Motion along curves; intersections; parametric differentiation in context.
<code>al.y2.pure.standard-derivatives</code>	Standard derivatives	Derivatives of $\sin$, $\cos$, $\tan$, e^x , a^x , $\ln x$.
<code>al.y2.pure.product-quotient-chain</code>	Product, quotient and chain rules	Differentiating products, quotients and compositions.
<code>al.y2.pure.implicit</code>	Implicit differentiation	Differentiating implicit relations; tangents and normals.
<code>al.y2.pure.parametric-differentiation</code>	Parametric differentiation	$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}}$; tangents and normals to parametric curves.
<code>al.y2.pure.connected-rates</code>	Rates of change	Connected rates via the chain rule; growth and decay.
<code>al.y2.pure.second-derivatives</code>	Second derivatives and curve behaviour	Convexity, concavity and points of inflection.
<code>al.y2.pure.roots-iteration</code>	Locating roots and iteration	Change of sign and its failures; fixed-point iteration; staircase/cobweb.

continued

id	subtopic	description
al.y2.pure.newton-raphson	Newton–Raphson	The method, its geometry and failure conditions.
al.y2.pure.numerical-integration	Numerical integration	The trapezium rule; over- and under-estimates.
al.y2.pure.standard-integrals	Standard integrals	Integrating e^x , $\frac{1}{x}$, trig functions and $f(ax + b)$.
al.y2.pure.integration-substitution	Integration by substitution	Substitution for definite and indefinite integrals, including trig.
al.y2.pure.integration-by-parts	Integration by parts	Single and repeated application, including $\int \ln x \, dx$.
al.y2.pure.integration-partial-fractions	Integration with partial fractions	Integrating rational functions after decomposition.
al.y2.pure.areas-applications	Areas and further applications	Areas between curves; parametric integration.
al.y2.pure.differential-equations	Differential equations	Separation of variables; general/particular solutions; growth and cooling.
al.y2.pure.vectors-3d	Vectors in three dimensions	3D coordinates, magnitude, distance; i, j, k problems.

11 Year 2 Statistics

id	subtopic	description
al.y2.stats.regression-correlation	Regression and correlation	Exponential models via logs; PMCC; test for zero correlation.
al.y2.stats.conditional-probability	Conditional probability	$P(A B)$; multiplication law; testing independence.
al.y2.stats.normal-distribution	The normal distribution	Normal curve; probabilities and inverse-normal; finding μ and σ .
al.y2.stats.normal-approximation	Normal approximation to binomial	Conditions and continuity corrections; model choice.
al.y2.stats.hypothesis-normal	Hypothesis testing (normal)	Tests for the mean using the distribution of the sample mean.

12 Year 2 Mechanics

id	subtopic	description
al.y2.mech.moments	Moments	Turning effects; equilibrium of rigid bodies; tilting; simple centres of mass.
al.y2.mech.forces-friction	Forces and friction	Resolving on inclined planes; $F \leq \mu R$ and limiting equilibrium.
al.y2.mech.projectiles	Projectiles	Horizontal and angled projection; range, height, trajectory formulae.

continued

id	subtopic	description
al.y2.mech. applications-forces	Applications of forces	Statics of particles; dynamics on slopes and with connected bodies.
al.y2.mech. further-kinematics	Further kinematics	Vector methods in 2D; variable acceleration via calculus of vectors.

Part IV — A-level Further Mathematics, Year 1

Key prefix fm.y1. — Core Pure 1 (pure), Further Statistics 1 (stats), Further Mechanics 1 (mech).

13 Further Maths Year 1: Pure (Core Pure 1)

id	subtopic	description
fm.y1.pure. complex-arithmetic	Complex numbers: arithmetic	Four operations; conjugates; quadratics with complex roots.
fm.y1.pure.argand	Argand diagrams	Representation; modulus and argument; mod-arg form; loci.
fm.y1.pure.roots-polynomials	Roots of polynomials	Root-coefficient relations for quadratics to quartics; transforming roots.
fm.y1.pure.induction	Proof by induction	Induction for series, divisibility and matrix powers.
fm.y1.pure.matrices	Matrices: arithmetic and determinants	Operations; 2×2 and 3×3 determinants and inverses; systems.
fm.y1.pure. linear-transformations	Linear transformations	Matrices for rotations, reflections, enlargements; invariant points/lines.
fm.y1.pure.series	Series	Standard results for $\sum r$, $\sum r^2$, $\sum r^3$ and related sums.
fm.y1.pure. vectors-lines-planes	Vectors: lines and planes	Vector/Cartesian lines and planes; scalar product; angles and distances.
fm.y1.pure. volumes-revolution	Volumes of revolution	Rotation about the x - and y -axes; composite regions; modelling.

14 Further Maths Year 1: Statistics (Further Statistics 1)

id	subtopic	description
fm.y1.stats.discrete-rv	Discrete random variables	Expectation and variance; $E(aX + b)$, $\text{Var}(aX + b)$; functions of X .
fm.y1.stats.poisson	The Poisson distribution	Conditions; mean and variance; additivity; approximation to binomial.
fm.y1.stats. geometric-negbinomial	Geometric and negative binomial	Trials to first and r th success; means and variances.
fm.y1.stats.hypothesis-tests	Hypothesis testing (Poisson, geometric)	Tests for Poisson and geometric parameters; critical regions.
fm.y1.stats.central-limit	Central limit theorem	Distribution of sample means for large samples; approximate inference.
fm.y1.stats.chi-squared	Chi-squared tests	Goodness of fit; contingency tables; degrees of freedom.

15 Further Maths Year 1: Mechanics (Further Mechanics 1)

id	subtopic	description
fm.y1.mech.momentum-impulse	Momentum and impulse	Impulse-momentum in 1D and 2D; conservation of momentum.

continued

id	subtopic	description
<i>fm.y1.mech.work-energy-power</i>	Work, energy and power	Work, KE, PE; work–energy principle; power on inclines.
<i>fm.y1.mech. elastic-strings-springs</i>	Elastic strings and springs	Hooke’s law; elastic PE; energy methods with elastic components.
<i>fm.y1.mech.collisions-1d</i>	Elastic collisions in one dimension	Newton’s law of restitution; direct impacts; successive collisions.

Part V — A-level Further Mathematics, Year 2

Key prefix fm.y2. — Core Pure 2 (pure), Further Statistics 2 (stats), Further Mechanics 2 (mech).

16 Further Maths Year 2: Pure (Core Pure 2)

id	subtopic	description
fm.y2.pure.de-moivre	Complex numbers: de Moivre	Exponential form; de Moivre's theorem; multiple-angle identities; trig sums.
fm.y2.pure.roots-of-unity	Roots of unity	n th roots of unity and of any complex number; geometric interpretation.
fm.y2.pure.method-of-differences	Series: method of differences	Telescoping sums, including with partial fractions.
fm.y2.pure.maclaurin	Maclaurin series	Series for e^x , $\ln(1+x)$, $\sin x$, $\cos x$, $(1+x)^n$; validity; composites.
fm.y2.pure.further-calculus	Further calculus	Improper integrals; mean value; inverse-trig integrals; partial fractions.
fm.y2.pure.polar-coordinates	Polar coordinates	Conversion; sketching polar curves; enclosed areas; tangents.
fm.y2.pure.hyperbolic-functions	Hyperbolic functions	$\sinh$, $\cosh$, $\tanh$ and inverses; identities; hyperbolic calculus.
fm.y2.pure.first-order-des	First-order differential equations	Integrating factors; solution-curve families; modelling.
fm.y2.pure.second-order-des	Second-order differential equations	Constant-coefficient linear equations; complementary function and PI.
fm.y2.pure.de-modelling	Modelling with differential equations	Simple harmonic motion; damped and forced oscillations; coupled systems.

17 Further Maths Year 2: Statistics (Further Statistics 2)

id	subtopic	description
fm.y2.stats.linear-regression	Linear regression	Least-squares model, residuals and RSS; interpreting fitted models.
fm.y2.stats.continuous-rv	Continuous random variables	PDFs and CDFs; expectation, variance, median, mode; functions of X .
fm.y2.stats.continuous-uniform	The continuous uniform distribution	Properties, moments and applications of the rectangular distribution.
fm.y2.stats.combining-rv	Combinations of random variables	Means and variances of linear combinations of independent variables.
fm.y2.stats.estimation-ci	Estimation and confidence intervals	Unbiased estimators; confidence intervals; the t -distribution.
fm.y2.stats.further-tests	Further hypothesis tests	Difference of means; F -test for variances; choosing a test.
fm.y2.stats.spearman	Spearman's rank correlation	Rank correlation coefficient and tests; when to prefer rank methods.

18 Further Maths Year 2: Mechanics (Further Mechanics 2)

id	subtopic	description
fm.y2.mech. horizontal-circular	Horizontal circular motion	Angular speed; $r\omega^2$; conical pendulum and banked tracks.
fm.y2.mech.vertical-circular	Vertical circular motion	Energy methods; conditions for complete circles.
fm.y2.mech.com-laminas	Centres of mass: laminas and frameworks	Uniform and composite laminas and frameworks by symmetry and moments.
fm.y2.mech.com-calculus	Centres of mass: solids and calculus	Standard solids and solids of revolution by integration.
fm.y2.mech.statics-rigid	Statics of rigid bodies	Equilibrium including suspension and toppling vs sliding.
fm.y2.mech.collisions-2d	Elastic collisions in two dimensions	Oblique impact with surfaces and spheres; restitution on the line of centres.
fm.y2.mech.further-dynamics	Further dynamics	Variable forces; $v \frac{dv}{dx}$ forms.